

Weighted Round Robin Arbiter - Complete Visual Report

This fixed landscape report includes the full description, neat FSM diagram, request-zero fallback behavior, EDA Playground steps, and the complete EPWave screenshot without cutting.

Design Summary

The arbiter has four request inputs, req0 to req3, and four one-hot grant outputs, master0 to master3. master0 has weight 4, while master1, master2, and master3 each have weight 1. Therefore, master0 receives four turns when all request signals are active.

Grant Sequence for Weight 4:1:1:1



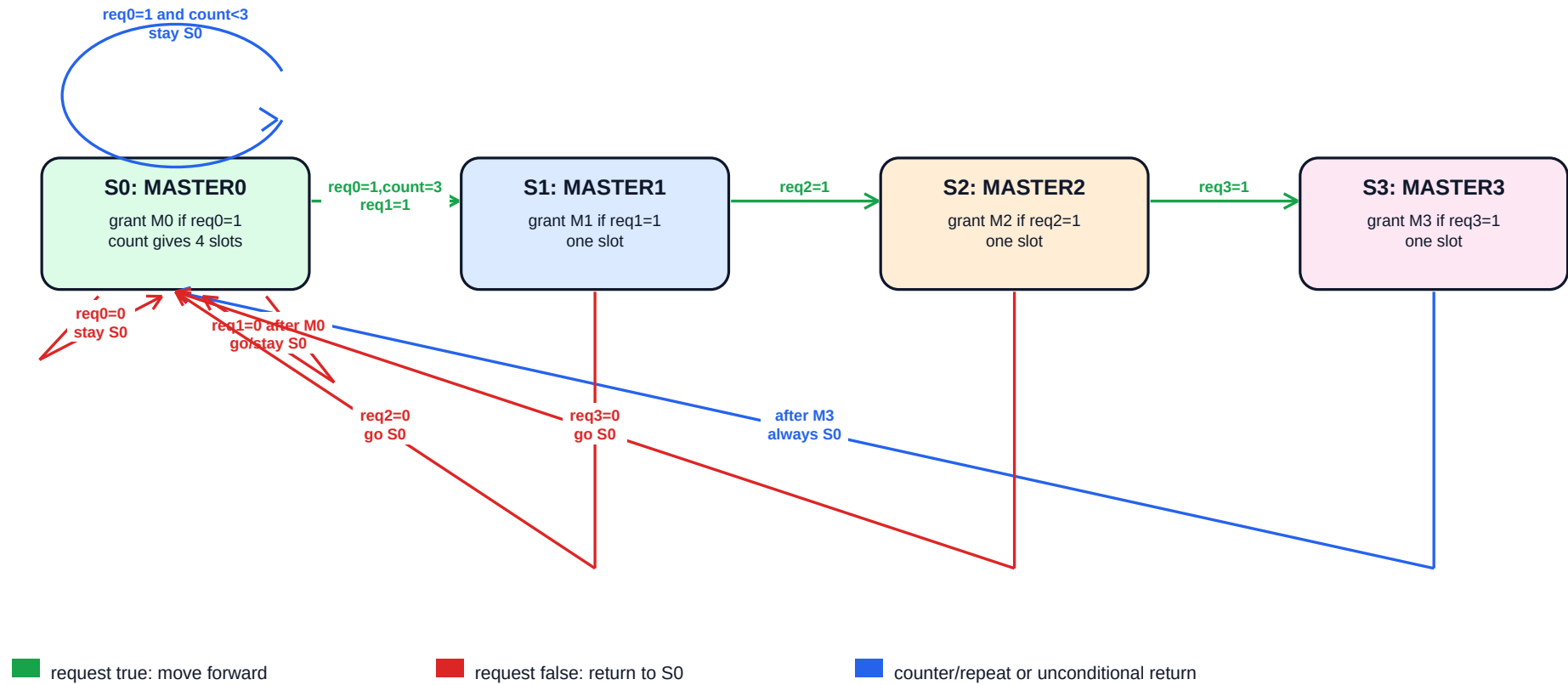
After M3, the cycle repeats from M0.

Grant sequence when all requests are high:

M0 -> M0 -> M0 -> M0 -> M1 -> M2 -> M3 -> repeat

FSM Diagram: Request 0 Arrows Return to S0

Green arrows: request is 1 and FSM moves forward. Red arrows: request is 0 and FSM goes back to S0.



FSM Request Conditions

Condition	FSM action
req0 = 0 in S0	Stay in S0, reset count, and do not grant master0.
req0 = 1 and count < 3 in S0	Stay in S0 and continue granting master0.
req1 = 1 after M0 finishes	Move from S0 to S1.
req1 = 0 after M0 finishes	Return or stay in S0.
req2 = 1 in S1	Move from S1 to S2.
req2 = 0 in S1	Return to S0.
req3 = 1 in S2	Move from S2 to S3.
req3 = 0 in S2	Return to S0.
After S3	Always return to S0.

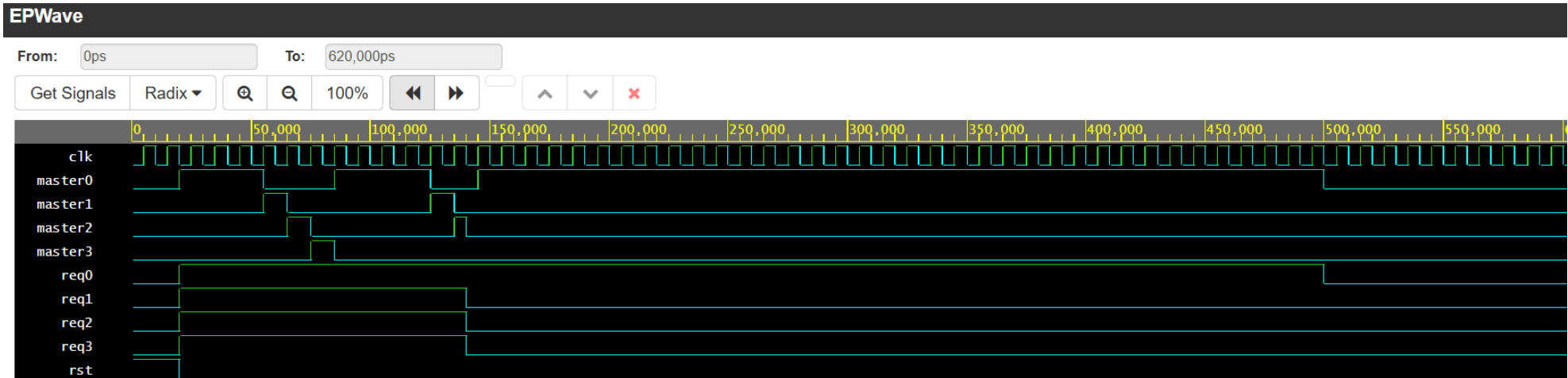
How to Run in EDA Playground

Paste arbiter.v in the design section and arbiter-tb.v in the testbench section. Select a Verilog simulator such as Icarus Verilog, then click Run. The testbench creates the waveform using:

```
$dumpfile("weighted_rr.vcd");  
$dumpvars(0,tb_weighted_rr_arbiter);
```

After simulation, open EPWave and view clk, rst, req0, req1, req2, req3, master0, master1, master2, and master3. If available, also add uut.state and uut.count.

EPWave Output



Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

Waveform Explanation

After reset goes low, the request inputs become active. master0 remains high for more clock cycles because it has weight 4. Then master1, master2, and master3 each receive one grant slot. Only one master output is high at a time, confirming one-hot arbitration.

Conclusion

The FSM and EPWave output show the complete weighted round robin behavior. Request-zero cases return to S0, request-one cases allow forward movement, and the counter in S0 gives master0 four grant slots before checking the next master.